

Theory of Design

Philosophy of Design

SEAtS | 080172

Intended Learning Outcomes

By the end of this lecture, you will be able to:

1. Define philosophy of design and distinguish its questions from everyday design practice questions.
2. Explain why the (conscious or unconscious) philosophies designers hold shape their practice.
3. Identify the key research questions in philosophy of design across the nature, epistemology, theory, and application of design.
4. Explain why rational and critical reflection—not empirical observation alone—is needed to answer these questions.
5. Evaluate the relevance of philosophy of design to design research, professional practice, and design education.

Agenda

- Design Shapes Our Lives
- What is Philosophy of Design?
- Key Research Questions
- Methodology
- Why It Matters
- Design as Shaping Reality
- Conclusion

Design Shapes Our Lives

Design shapes our lives as much as science and technology do—through countless artefacts we use daily. But design cannot be understood through science and technology alone.

Understanding design requires serious philosophical work.

Philosophy of design is a young field combining philosophy and design research. Most philosophers of design are "half amateurs"—they come from mixed backgrounds (philosophy, design practice, design research). This cross-disciplinary nature is essential.

What is Philosophy of Design?

Definition

Philosophy of design = the pursuit of insights about design through philosophical means

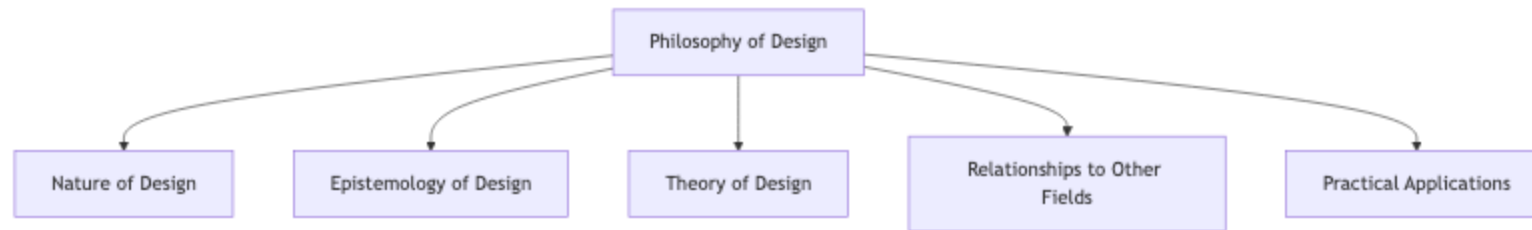
It addresses fundamental, abstract questions about design—more general than everyday design practice questions.

Core Principle

Our philosophies (conscious or unconscious) shape our actions. Design philosophy's impact on practice can be **devastating** if poorly understood.

Improving design requires improving the philosophy underlying design thinking.

Key Research Questions in Philosophy of Design



Five clusters of questions structure the field—from what design *is*, to how we can *know* about designed things, to what design theory should look like, and what use any of this is.

Nature of Design

- What distinguishes design from other intellectual endeavors (science, art, etc.)?
- How are "design" and "artefact" conceptually related?
- Is it necessary, sufficient, or both, that something was designed to be an artefact?
- Can there be artefacts without design? Can there be design without artefacts?

Can there be artefacts without design? Can there be design without artefacts?



4.00 **Turn and Talk**

Epistemology of Design

- How do designers talk meaningfully about artefacts that don't yet exist?
- Are statements of design discourse true, false, or meaningful? What makes them so?
- Can designers know or predict properties of artefacts not yet constructed?
- What ontological and epistemological assumptions enable this?

Theory of Design

- How should universals (properties/concepts) relate to design theory?
- Are there nominalist, conceptualist, and realist theories of design (mirroring philosophy of universals)? (HOMEWORK)
- What would design theories of each type look like, and what are their strengths/weaknesses?

Relationships to Other Fields

- How does philosophy of design relate to general philosophy?
- Are central design problems special cases of familiar philosophical problems, or new ones?
- Can philosophy of design offer insights to other philosophical fields?
- Can other fields of philosophy illuminate design problems?

Practical Applications

- Can philosophy of design improve design practice?
- Can it lead to better artefacts, better design methods, better use of computers in design?
- What use can designers make of philosophical insights?

Methodology: How Philosophy of Design Works

Philosophy of design uses:

- **Rational and critical reflection** (not empirical observation alone)
- Conceptual analysis and argumentative power
- Methods that transcend empirical research
- Cultivation of awareness about fundamental assumptions

Why This Matters

Empirical design research (protocol analysis, user studies, etc.) cannot answer these fundamental questions.

Protocol analysis won't tell you whether design is possible without artefacts—that requires philosophical reflection on the nature of design itself.

Motivation



Why Philosophy of Design Matters

For Design Research

- Provides theoretical foundations for understanding what design is
- Clarifies assumptions underlying research methods
- Connects design research to philosophical traditions

For Professional Practice

- Can improve design methods and processes
- Can lead to better artefacts and outcomes
- Can improve how computers and tools are used in design
- Raises awareness of implicit assumptions that shape practice

For Design Education

- Shapes how designers think about their discipline
- Cultivates philosophical awareness in future designers
- Connects design to broader intellectual traditions
- Improves the quality of design thinking

Design as Shaping Reality

Design is not merely technical problem-solving. It is:

- A philosophical endeavor involving fundamental questions about reality, knowledge, and purpose
- Shaped by implicit philosophies that drive design decisions
- A practice that requires critical reflection on its own nature and assumptions
- Essential to understanding how artefacts shape human life and environment

Just as science requires philosophy of science, design requires philosophy of design to be fully understood and practised well.

Conclusion

Philosophy of design is a young, cross-disciplinary field pursuing insights about design through philosophical—not purely empirical—means.

Its questions span the nature of design, the epistemology of talking about artefacts that don't yet exist, the theory of design universals, design's relationship to other fields, and its practical payoff.

Because our implicit philosophies already shape how we design, making them explicit is not an academic luxury: it is how design research, practice, and education improve.

Thank you

Dr Nawfal Al Hashimy

Lecturer in Cyber Physical Security

University of Southampton | Faculty of Engineering and Physical
Sciences

Cyber Physical Systems Research Group

☎ +44 (0) 23 8059 6659

@ Nawfal@soton.ac.uk

Highfield Campus, Building 53, Room 4049
Burgess Road, Southampton SO17 1BJ, UK

